

Galaxy

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*** START OF THE PROJECT GUTENBERG EBOOK BIG ANCESTOR

BIG ANCESTOR

By F. L. WALLACE

Illustrated by EMSH

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Man's family tree was awesome enough to give every galactic race an inferiority complex—but then he tried to climb it!

In repose, Taphetta the Ribboneer resembled a fancy giant bow on a package. His four flat legs looped out and in, the ends tucked under his wide, thin body, which constituted the knot at the middle. His neck was flat, too, arching out in another loop. Of all his features, only his head had appreciable thickness and it was crowned with a dozen long though narrower ribbons.

Taphetta rattled the head fronds together in a surprisingly good imitation of speech. "Yes, I've heard the legend."

"It's more than a legend," said Sam Halden, biologist. The reaction was not unexpected—non-humans tended to dismiss the data as convenient speculation and nothing more. "There are at least a hundred kinds of humans, each supposedly originating in strict seclusion on as many widely scattered planets. Obviously there was no contact throughout the ages before space travel—and yet *each planetary race can interbreed with a minimum of ten others!* That's more than a legend—one hell of a lot more!"

"It is impressive," admitted Taphetta. "But I find it mildly distasteful to consider mating with someone who does not belong to my species."

"That's because you're unique," said Halden. "Outside of your own world, there's nothing like your species, except superficially, and that's true of all other creatures, intelligent or not, with the sole exception of mankind. Actually, the four of us here, though it's accidental, very nearly represent the biological spectrum of human development."



"Emmer, a Neanderthal type and our archeologist, is around the beginning of the scale. I'm from Earth, near the middle, though on Emmer's side. Meredith, linguist, is on the other side of the middle. And beyond her, toward the far end, is Kelburn, mathematician. There's a corresponding span of fertility. Emmer just misses being able to breed with my kind, but there's a fair chance that I'd be fertile with Meredith and a similar though lesser chance that her fertility may extend to Kelburn."

Taphetta rustled his speech ribbons quizzically. "But I thought it was proved that some humans did originate on one planet, that there was an unbroken line of evolution that could be traced back a billion years."

"You're thinking of Earth," said Halden. "Humans require a certain kind of planet. It's reasonable to assume that, if men were set down on a hundred such worlds, they'd seem to fit in with native life-forms on a few of them. That's what happened on Earth; when Man arrived, there was actually a manlike creature there. Naturally our early evolutionists stretched their theories to cover the facts they had."

"But there are other worlds in which humans who were there before the Stone Age aren't related to anything else there. We have to conclude that Man didn't originate on any of the planets on which he is now found. Instead, he evolved elsewhere and later was scattered throughout this section of the Milky Way."

"And so, to account for the unique race that can interbreed across thousands of light-years, you've brought in the big ancestor," commented Taphetta dryly. "It seems an unnecessary simplification."

"Can you think of a better explanation?" asked Kelburn.

"Something had to distribute one species so widely and it's not the result of parallel evolution—not when a hundred human races are involved, and *only* the human race."

"I can't think of a better explanation." Taphetta rearranged his ribbons. "Frankly, no one else is much interested in Man's theories about himself."

It was easy to understand the attitude. Man was the most numerous though not always the most advanced—Ribboneers had a civilization as high as anything in the known section of the Milky Way, and there were others—and humans were more than a little feared. If they ever got together—but they hadn't except in agreement as to their common origin.

Still, Taphetta the Ribboneer was an experienced pilot and could be very useful. A clear statement of their position was essential in helping him make up his mind. "You've heard of the adjacency mating principle?" asked Sam Halden.

"Vaguely. Most people have if they've been around men."

"We've got new data and are able to interpret it better. The theory is that humans who can mate with each other were once physically close. We've got a list of all our races arranged in sequence. If planetary race F can mate with race E back to A and forward to M, and race G is fertile only back to B, but forward to O, then we assume that whatever their positions are now, at once time G was actually adjacent to F, but was a little further along. When we project back into time those star systems on which humans existed prior to space travel, we get a certain pattern. Kelburn can explain it to you."

The normally pink body of the Ribboneer flushed slightly. The color change was almost imperceptible, but it was enough to indicate that he was interested.

Kelburn went to the projector. "It would be easier if we knew all the stars in the Milky Way, but though we've explored only a small portion of it, we can reconstruct a fairly accurate representation of the past."

He pressed the controls and stars twinkled on the screen. "We're looking down on the plane of the Galaxy. This is one arm of it as it is today and here are the human systems." He pressed another control and, for purposes of identification, certain stars became more brilliant. There was no pattern, merely a scattering of stars. "The whole Milky Way is rotating. And while stars in a given region tend to remain together, there's also a random motion. Here's what happens when we calculate the positions of stars in the past."

Flecks of light shifted and flowed across the screen. Kelburn stopped the motion.

"Two hundred thousand years ago," he said.

There was a pattern of the identified stars. They were spaced at fairly equal intervals along a regular curve, a horseshoe loop that didn't close, though if the ends were extended, the lines would have crossed.

Taphetta rustled. "The math is accurate?"

"As accurate as it can be with a million-plus body problem."

"And that's the hypothetical route of the unknown ancestor?"

"To the best of our knowledge," said Kelburn. "And whereas there are humans who are relatively near and not fertile, they can always mate with those they were adjacent to *two hundred thousand years ago!*"

"The adjacency mating principle. I've never seen it demonstrated," murmured Taphetta, flexing his ribbons. "Is that the only era that satisfies the calculations?"

"Plus or minus a hundred thousand years, we can still get something that might be the path of a spaceship attempting to cover a representative section of territory," said Kelburn. "However, we have other ways of dating it. On some worlds on which there are no other mammals, we're able to place the first human fossils chronologically. The evidence is sometimes contradictory, but we believe we've got the time right."

Taphetta waved a ribbon at the chart. "And you think that where the two ends of the curve cross is your original home?"

"We think so," said Kelburn. "We've narrowed it down to several cubic light-years—then. Now it's far more. And, of course, if it were a fast-moving star, it might be completely out of the field of our exploration. But we're certain we've got a good chance of finding it this trip."

"It seems I must decide quickly." The Ribboneer glanced out the visionport, where another ship hung motionless in space beside them. "Do you mind if I ask other questions?"

"Go ahead," Kelburn invited sardonically. "But if it's not math, you'd better ask Halden. He's the leader of the expedition."

Halden flushed; the sarcasm wasn't necessary. It was true that Kelburn was the most advanced human type present, but while there were differences, biological and in the scale of intelligence, it wasn't as great as once was thought. Anyway, non-humans weren't trained in the fine distinctions that men made among themselves. And, higher or lower, he was as good a biologist as the other was a mathematician. And there was the matter of training; he'd been on several expeditions and this was Kelburn's first trip. Damn it, he thought, that rated some respect.

The Ribboneer shifted his attention. "Aside from the sudden illness of your pilot, why did you ask for me?"

"We didn't. The man became sick and required treatment we can't give him. Luckily, a ship was passing and we hailed it because it's four months to the nearest planet. They consented to take him back and told us that there was a passenger on board who was an experienced pilot. We have men who could do the job in a makeshift fashion, but the region we're heading for, while mapped, is largely unknown. We'd prefer to have an expert—and Ribboneers are famous for their navigational ability."

Taphetta crinkled politely at the reference to his skill. "I had other plans, but I can't evade professional obligations, and an emergency such as this should cancel out any previous agreements. Still, what are the incentives?"

Sam Halden coughed. "The usual, plus a little extra. We've copied the Ribboneer's standard nature, simplifying it a little and adding a per cent here and there for the crew pilot and scientist's share of the profits from any discoveries we may make."

"I'm complimented that you like our contract so well," said Taphetta, "but I really must have our own unsimplified version. If you want me, you'll take my contract. I came prepared." He extended a tightly bound roll that he had kept somewhere on his person.

They glanced at one another as Halden took it.

"You can read it if you want," offered Taphetta. "But it will take you all day—it's micro-printing. However, you needn't be afraid that I'm defrauding you. It's honored everywhere we go and we go nearly everywhere in this sector—places men have never been."

There was no choice if they wanted him, and they did. Besides, the integrity of Ribboneers was not to be questioned. Halden signed.

"Good." Taphetta crinkled. "Send it to the ship; they'll forward it for me. And you can tell the ship to go on without me." He rubbed his ribbons together. "Now if you'll get me the charts, I'll examine the region toward which we're heading."

Firmon of hydroponics slouched in, a tall man with scanty hair and an equal lack of grace. He seemed to have difficulty in taking his eyes off Meredith, though, since he was a notch or so above her in the mating scale, he shouldn't have been so interested. But his planet had been inexplicably slow in developing and he wasn't completely aware of his place in the human hierarchy.

Disdainfully, Meredith adjusted a skirt that, a few inches shorter, wouldn't have been a skirt at all, revealing, while doing so, just how long and

beautiful a woman's legs could be. Her people had never given much thought to physical modesty and, with legs like that, it was easy to see why.

Muttering something about primitive women, Firmon turned to the biologist. "The pilot doesn't like our air."

"Then change it to suit him. He's in charge of the ship and knows more about these things than I do."

"More than a man?" Firmon leered at Meredith and, when she failed to smile, added plaintively, "I did try to change it, but he still complains."

Halden took a deep breath. "Seems all right to me."

"To everybody else, too, but the tapeworm hasn't got lungs. He breathes through a million tubes scattered over his body."

It would do no good to explain that Taphetta wasn't a worm, that his evolution had taken a different course, but that he was in no sense less complex than Man. It was a paradox that some biologically higher humans hadn't developed as much as lower races and actually weren't prepared for the multitude of life-forms they'd meet in space. Firmon's reaction was quite typical.

"If he asks for cleaner air, it's because his system needs it," said Halden. "Do anything you can to give it to him."

"Can't. This is as good as I can get it. Taphetta thought you could do something about it."

"Hydroponics is your job. There's nothing *I* can do." Halden paused thoughtfully. "Is there something wrong with the plants?"

"In a way, I guess, and yet not really."

"What is it, some kind of toxic condition?"

"The plants are healthy enough, but something's chewing them down as fast as they grow."

"Insects? There shouldn't be any, but if there are, we've got sprays. Use them."

"It's an animal," said Firmon. "We tried poison and got a few, but now they won't touch the stuff. I had electronics rig up some traps. The animals seem to know what they are and we've never caught one that way."

Halden glowered at the man. "How long has this been going on?"

"About three months. It's not bad; we can keep up with them."

It was probably nothing to become alarmed at, but an animal on the ship was a nuisance, doubly so because of their pilot.

"Tell me what you know about it," said Halden.

"They're little things." Firmon held out his hands to show how small. "I don't know how they got on, but once they did, there were plenty of places to hide." He looked up defensively. "This is an old ship with new equipment and they hide under the machinery. There's nothing we can do except rebuild the ship from the hull inward."

Firmon was right. The new equipment had been installed in any place just to get it in and now there were inaccessible corners and crevices everywhere that couldn't be closed off without rebuilding.

They couldn't set up a continuous watch and shoot the animals down because there weren't that many men to spare. Besides, the use of weapons in hydroponics would cause more damage to the thing they were trying to protect than to the pest. He'd have to devise other ways.

Sam Halden got up. "I'll take a look and see what I can do."

"I'll come along and help," said Meredith, untwining her legs and leaning against him. "Your mistress ought to have some sort of privileges."

Halden started. So she *knew* that the crew was calling her that! Perhaps it was intended to discourage Firmon, but he wished she hadn't said it. It didn't help the situation at all.

Taphetta sat in a chair designed for humans. With a less flexible body, he wouldn't have fitted. Maybe it wasn't sitting, but his flat legs were folded neatly around the arms and his head rested comfortably on the seat. The head ribbons, which were his hands and voice, were never quite still.

He looked from Halden to Emmer and back again. "The hydroponics tech tells me you're contemplating an experiment. I don't like it."

Halden shrugged. "We've got to have better air. It might work."

"Pests on the ship? It's filthy! My people would never tolerate it!"

"Neither do we."

The Ribboneer's distaste subsided. "What kind of creatures are they?"

"I have a description, though I've never seen one. It's a small four-legged animal with two antennae at the lower base of its skull. A typical pest."

Taphetta rustled. "Have you found out how it got on?"

"It was probably brought in with the supplies," said the biologist. "Considering how far we've come, it may have been any one of a half a dozen planets. Anyway, it hid, and since most of the places it had access to were near the outer hull, it got an extra dose of hard radiation, or it may have nested near the atomic engines; both are possibilities. Either way, it mutated, became a different animal. It's developed a tolerance for the poisons we spray on plants. Other things it detects and avoids, even electronic traps."

"Then you believe it changed mentally as well as physically, that it's smarter?"

"I'd say that, yes. It must be a fairly intelligent creature to be so hard to get rid of. But it can be lured into traps, if the bait's strong enough."

"That's what I don't like," said Taphetta, curling. "Let me think it over while I ask questions." He turned to Emmer. "I'm curious about humans. Is there anything else you can tell me about the hypothetical ancestor?"

Emmer didn't look like the genius he was—a Neanderthal genius, but nonetheless a real one. In his field, he rated very high. He raised a stubble-flecked cheek from a large thick-fingered paw and ran shaggy hands through shaggier hair.

"I can speak with some authority," he rumbled. "I was born on a world with the most extensive relics. As a child, I played in the ruins of their camp."

"I don't question your authority," crinkled Taphetta. "To me, all humans—late or early and male or female—look remarkably alike. If you are an archeologist, that's enough for me." He paused and flicked his speech ribbons. "Camp, did you say?"

Emmer smiled, unsheathing great teeth. "You've never seen any pictures? Impressive, but just a camp, monolithic one-story structures, and we'd give something to know what they're made of. Presumably my world was one of the first they stopped at. They weren't used to roughing it, so they built more elaborately than they did later on. One-story structures and that's how we can guess at their size. The doorways were forty feet high."

"Very large," agreed Taphetta. It was difficult to tell whether he was impressed. "What did you find in the ruins?"

"Nothing," said Emmer. "There were buildings there and that was all, not a scrap of writing or a tool or a single picture. They covered a route estimated at thirty thousand light-years in less than five thousand years—and not one of them died that we have a record of."

"A faster-than-light drive and an extremely long life," mused Taphetta. "But they didn't leave any information for their descendants. Why?"

"Who knows? Their mental processes were certainly far different from ours. They may have thought we'd be better off without it. We do know they were looking for a special kind of planet, like Earth, because they visited so many of that type, yet different from it because they never stayed. They were pretty special people themselves, big and long-lived, and maybe they couldn't survive on any planet they found. Perhaps they had ways of determining there wasn't the kind of planet they needed in the entire Milky Way. Their science was tremendously advanced and when they learned that, they may have altered their germ plasm and left us, hoping that some of us would survive. Most of us did."

"This special planet sounds strange," murmured Taphetta.

"Not really," said Emmer. "Fifty human races reached space travel independently and those who did were scattered equally among early and late species. It's well known that individuals among my people are often as bright as any of Halden's or Meredith's, but as a whole we don't have the total capacity that later Man does, and yet we're as advanced in civilization. The difference? It must lie somewhere in the planets we live on and it's hard to say just what it is."

"What happened to those who didn't develop space travel?" asked Taphetta.

"We helped them," said Emmer.

And they had, no matter who or what they were, biologically late or early, in the depths of the bronze age or the threshold of atomic—because they were human. That was sometimes a frightening thing for non-humans, that the race stuck together. They weren't actually aggressive, but their total number was great and they held themselves aloof. The unknown ancestor again. Who else had such an origin and, it was tacitly assumed, such a destiny?

Taphetta changed his questioning. "What do you expect to gain from this discovery of the unknown ancestor?"

It was Halden who answered him. "There's the satisfaction of knowing where we came from."

"Of course," rustled the Ribboneer. "But a lot of money and equipment was required for this expedition. I can't believe that the educational institutions that are backing you did so purely out of intellectual curiosity."

"Cultural discoveries," rumbled Emmer. "How did our ancestors live? When a creature is greatly reduced in size, as we are, more than physiology is changed—the pattern of life itself is altered. Things that were easy for them are impossible for us. Look at their life span."

"No doubt," said Taphetta. "An archeologist would be interested in cultural discoveries."

"Two hundred thousand years ago, they had an extremely advanced civilization," added Halden. "A faster-than-light drive, and we've achieved that only within the last thousand years."

"But I think we have a better one than they did," said the Ribboneer. "There may be things we can learn from them in mechanics or physics, but wouldn't you say they were better biologists than anything else?"

Halden nodded. "Agreed. They couldn't find a suitable planet. So, working directly with their germ plasm, they modified themselves and produced us. They *were* master biologists."

"I thought so," said Taphetta. "I never paid much attention to your fantastic theories before I signed to pilot this ship, but you've built up a convincing case." He raised his head, speech ribbons curling fractionally and ceaselessly. "I don't like to, but we'll have to risk using bait for your pest."

He'd have done it anyway, but it was better to have the pilot's consent. And there was one question Halden wanted to ask; it had been bothering him vaguely. "What's the difference between the Ribboneer contract and the one we offered you? Our terms are more liberal."

"To the individual, they are, but it won't matter if you discover as much as you think you will. The difference is this: *My* terms don't permit you to withhold any discovery for the benefit of one race."

Taphetta was wrong; there had been no intention of withholding anything. Halden examined his own attitudes. *He* hadn't intended, but could he say that was true of the institutions backing the expedition? He couldn't, and it was too late now—whatever knowledge they acquired would have to be shared.

That was what Taphetta had been afraid of—there was one kind of technical advancement that multiplied unceasingly. The race that could improve itself through scientific control of its germ plasm had a start that could never be headed. The Ribboneer needn't worry now.

"Why do we have to watch it on the screen?" asked Meredith, glancing up. "I'd rather be in hydroponics."

Halden shrugged. "They may or may not be smarter than planetbound animals, but they're warier. They don't come out when anyone's near."

Lights dimmed in the distant hydroponic section and the screen with it, until he adjusted the infra-red frequencies. He motioned to the two crew members, each with his own peculiar screen, below which was a miniature keyboard.

"Ready?"

When they nodded, Halden said: "Do as you've rehearsed. Keep noise at a minimum, but when you do use it, be vague. Don't try to imitate them exactly."

At first, nothing happened on the big screen, and then a gray shape crept out. It slid through leaves, listened intently before coming forward. It jumped off one hydroponic section and fled across the open floor to the next. It paused, eyes glittering and antennae twitching.

Looking around once, it leaped up, seizing the ledge and clawing up the side of the tank. Standing on top and rising to its haunches, it began nibbling what it could reach.

Suddenly it whirled. Behind it and hitherto unnoticed was another shape, like it but larger. The newcomer inched forward. The small one retreated, skittering nervously. Without warning, the big one leaped and the small one tried to flee. In a few jumps, the big one caught up and mauled the other unmercifully.



It continued to bite even after the little one lay still. At last it backed off and waited, watching for signs of motion. There was none. Then it turned to the plant. When it had chewed off everything within reach, it climbed into the branches.

The little one twitched, moved a leg, and cautiously began dragging itself away. It rolled off the raised section and surprisingly made no noise as it fell. It seemed to revive, shaking itself and scurrying away, still within range of the screen.

Against the wall was a small platform. The little one climbed on top and there found something that seemed to interest it. It sniffed around and

reached and felt the discovery. Wounds were forgotten as it snatched up the object and frisked back to the scene of its recent defeat.

This time it had no trouble with the raised section. It leaped and landed on top and made considerable noise in doing so. The big animal heard and twisted around. It saw and clambered down hastily, jumping the last few feet. Squealing, it hit the floor and charged.

The small one stood still till the last instant—and then a paw flickered out and an inch-long knife blade plunged into the throat of the charging creature. Red spurted out as the bigger beast screamed. The knife flashed in and out until the big animal collapsed and stopped moving.

The small creature removed the knife and wiped it on the pelt of its foe. Then it scampered back to the platform on which the knife had been found—and *laid it down*.

At Halden's signal, the lights flared up and the screen became too bright for anything to be visible.

"Go in and get them," said Halden. "We don't want the pests to find out that the bodies aren't flesh."

"It was realistic enough," said Meredith as the crewmen shut off their machines and went out. "Do you think it will work?"

"It might. We had an audience."

"Did we? I didn't notice." Meredith leaned back. "Were the puppets exactly like the pests? And if not, will the pests be fooled?"

"The electronic puppets were a good imitation, but the animals don't have to identify them as their species. If they're smart enough, they'll know the value of a knife, no matter who uses it."

"What if they're smarter? Suppose they know a knife can't be used by a creature without real hands?"

"That's part of our precautions. They'll never know until they try—and they'll never get away from the trap to try."

"Very good. I never thought of that," said Meredith, coming closer. "I like the way your primitive mind works. At times I actually think of marrying you."

"Primitive," he said, alternately frozen and thawed, though he knew that, in relation to her, he was *not* advanced.

"It's almost a curse, isn't it?" She laughed and took the curse away by leaning provocatively against him. "But barbaric lovers are often nice."

Here we go again, he thought drearily, sliding his arm around her. To her, I'm merely a passionate savage.

They went to his cabin.

She sat down, smiling. Was she pretty? Maybe. For her own race, she wasn't tall, only by Terran standards. Her legs were disproportionately long and well shaped and her face was somewhat bland and featureless, except for a thin, straight, short nose. It was her eyes that made the difference, he decided. A notch or two up the scale of visual development, her eyes were larger and she could see an extra color on the violet end of the spectrum.

She settled back and looked at him. "It might be fun living with you on primeval Earth."

He said nothing; she knew as well as he that Earth was as advanced as her own world. She had something else in mind.

"I don't think I will, though. We might have children."

"Would it be wrong?" he asked. "I'm as intelligent as you. We wouldn't have subhuman monsters."

"It would be a step up—for you." Under her calm, there was tension. It had been there as long as he'd known her, but it was closer to the surface now. "Do I have the right to condemn the unborn? Should I make them start lower than I am?"

The conflict was not new nor confined to them. In one form or another, it governed personal relations between races that were united against non-humans, but held sharp distinctions themselves.

"I haven't asked you to marry me," he said bluntly.

"Because you're afraid I'd refuse."

It was true; no one asked a member of a higher race to enter a permanent union.

"Why did you ever have anything to do with me?" demanded Halden.

"Love," she said gloomily. "Physical attraction. But I can't let it lead me astray."

"Why not make a play for Kelburn? If you're going to be scientific about it, he'd give you children of the higher type."

"Kelburn." It didn't sound like a name, the way she said it. "I don't like him and he wouldn't marry me."

"He wouldn't, but he'd give you children if you were humble enough. There's a fifty per cent chance you might conceive."

She provocatively arched her back. Not even the women of Kelburn's race had a body like hers and she knew it.

"Racially, there should be a chance," she said. "Actually, Kelburn and I would be infertile."

"Can you be sure?" he asked, knowing it was a poor attempt to act unconcerned.

"How can anyone be sure on a theoretical basis?" she asked, an oblique smile narrowing her eyes. "I know we can't."

His face felt anesthetized. "Did you have to tell me that?"

She got up and came to him. She nuzzled against him and his reaction was purely reflexive. His hand swung out and he could feel the flesh give when his knuckles struck it.

She fell back and dazedly covered her face with her hand. When she took it away, blood spurted. She groped toward the mirror and stood in front of it. She wiped the blood off, examining her features carefully.

"You've broken my nose," she said factually. "I'll have to stop the blood and pain."

She pushed her nose back into place and waggled it to make sure. She closed her eyes and stood silent and motionless. Then she stepped back and looked at herself critically.

"It's set and partially knitted. I'll concentrate tonight and have it healed by morning."

She felt in the cabinet and attached an invisible strip firmly across the bridge. Then she came over to him.

"I wondered what you'd do. You didn't disappoint me."

He scowled miserably at her. Her face was almost plain and the bandage, invisible or not, didn't improve her appearance any. How could he still feel that attraction to her?

"Try Emmer," he suggested tiredly. "He'll find you irresistible, and he's even more savage than I am."

"Is he?" She smiled enigmatically. "Maybe, in a biological sense. Too much, though. You're just right."

He sat down on the bed. Again there was only one way of knowing what Emmer would do—and she knew. She had no concept of love outside of the physical, to make use of her body so as to gain an advantage—what advantage?—for the children she intended to have. Outside of that, nothing mattered, and for the sake of alloying the lower with the higher, she was as cruel to herself as she was to him. And yet he wanted her.

"I do think I love you," she said. "And if love's enough, I may marry you in spite of everything. But you'll have to watch out whose children I have." She wriggled into his arms.

The racial disparity was great and she had provoked him, but it was not completely her fault. Besides....

Besides what? She had a beautiful body that could bear superior children—and they might be his.

He twisted away. With those thoughts, he was as bad as she was. Were they all that way, every one of them, crawling upward out of the slime toward

the highest goal they could conceive of? Climbing over—no, *through*—everybody they could coerce, seduce or marry—onward and upward. He raised his hand, but it was against himself that his anger was turned.

"Careful of the nose," she said, pressing against him. "You've already broken it once."

He kissed her with sudden passion that even he knew was primitive.

There were no immediate results from the puppet performance and so it was repeated at intervals. After the third time, Firmon reported, coming in as Halden pored over the meager biological data he'd gathered on the unknown ancestor. Wild guesses mostly, not one real fact in all the statistics. After two hundred thousand years, there wasn't much left to work with.

Firmon slouched down. "It worked," he said. "Got three a few hours ago."

Halden looked at him; he had hoped it wouldn't work. There was satisfaction in being right, but he would rather face something less intelligent. Wariness was one thing, the shyness and slyness of an unseen animal, but intelligence was more difficult to predict.

"Where are they?" he asked.

"Did you want them?" Firmon seemed surprised at the idea.

Halden sighed; it was his own fault. Firmon had a potentially good mind, but he hadn't been trained to use it and that counted for more than people thought. "Any animal smart enough to appreciate the value of a knife is worth study on that account. That goes double when it's a pest."

"I'll change the cremation setting," said Firmon. "Next time, we'll just stun them."

The trap setting was changed and several animals were taken. Physically, they were very much as Halden had described them to Taphetta, small four-legged creatures with fleshy antennae. Dissection revealed a fairly large brain capacity, while behavior tests indicated an intelligence somewhat

below what he had assumed. Still, it was more than he wanted a pest to have, especially since it also had hands.

The biological mechanism of the hands was simple. It walked on the back of the front paws, on the fingers of which were fleshy pads. When it sat upright, as it often did, the flexibility of the wrists permitted the forepaws to be used as hands. Clumsy, but because it had a thumb, it could handle such tools as a knife.

He had made an error there. He had guessed the intelligence, but he hadn't known it could use the weapon he had put within reach. A tiny thing with an inch-long knife was not much more dangerous than the animal alone, but he didn't like the idea of it loose on the ship.

The metal knife would have to be replaced with something else. Technicians could compound a plastic that would take a keen edge for a while and deteriorate to a soft mass in a matter of weeks. Meanwhile, he had actually given the animal a dangerous weapon—the concept of a tool. There was only one way to take that away from them, by extermination. But that would have to wait.

Fortunately, the creature had a short life and a shorter breeding period. The actual replacement rate was almost negligible. In attaining intelligence, it had been short-changed in fertility and, as a consequence, only in the specialized environment of this particular ship was it any menace at all.

They were lucky; a slightly higher fertility and the thing could threaten their existence. As it was, the ship would have to be deverminized before it could land on an inhabited planet.

Halden took the data to the Ribboneer pilot and, after some discussion, it was agreed that the plastic knife should supplant the metal one. It was also decided to allow a few to escape with the weapon; there had to be some incentive if the creature was to visit the trap more than a few times. Besides, with weapons there was always the chance of warfare between different groups. They might even exterminate each other.

Gradually, over a period of weeks, the damage to hydroponics subsided; the pests were under control. There was nothing to worry about unless they mutated again, which was unlikely.

Kelburn scowled at the pilot. "Where are we now?" he challenged, his face creased with suspicion.

"You have access to all the instruments, so you should know," said Taphetta. He was crouching and seemed about to spring, but he was merely breathing relaxedly through a million air tubes.

"I do know. My calculations show one star as the most probable. We should have reached it two days ago—and we're nowhere near it."

"True," admitted Taphetta. "We're heading toward what you would consider the fifth or sixth most likely star."

Kelburn caught the implication. They all did. "Then you know where it is?" he asked, suspicion vanishing.

"Not in the sense you're asking—no, I'm not sure it's what you're looking for. But there was once a great civilization there."

"You knew this and didn't tell us?"

"Why should I?" Taphetta looked at him in mild astonishment. "Before you hired me, I wouldn't tell you for obvious reasons. And afterward—well, you engaged all my skill and knowledge and I used them to bring you here by the shortest route. I didn't think it necessary to tell you until we actually arrived. Is that wrong?"

It wasn't wrong; it merely illustrated the difference in the way an alien mind worked. Sooner or later, they would have found the place, but he had saved them months.

"What's it like?" Emmer asked.

Taphetta juggled his ribbons. "I don't know. I was passing near here and saw the planet off to one side."

"And you didn't stop?" Emmer was incredulous.

"Why should I? We're great navigators because we do so much of it. We would never get very far if we stopped to examine everything that looks interesting. Besides, it's not a good policy in a strange region, especially with an unarmed ship."

They wouldn't have that problem. The ship was armed well enough to keep off uncivilized marauders who had very recently reached the spaceship age, and only such people were apt to be inhospitable.

"When will we land?" asked Halden.

"In a few hours, but you can see the planet on our screens." Taphetta extended a head ribbon toward a knob and a planet came into view.

There weren't two civilizations in the Milky Way that built on such a large scale, even from the distance that they could see it. Great, distinctive cities were everywhere. There was no question as to what they had found.

"Now you'll learn why they ran away," said Taphetta.

"A new theory," Kelburn said, though it wasn't, for they *had* left. "What makes you think they were afraid?"

"No air. If your calculations are right, there must have been an extensive atmosphere a few hundred thousand years ago and now there isn't any. A planet this size doesn't lose air that fast. Therefore, it's an artificial condition. Who takes the trouble to leave a planet uninhabitable except someone who's afraid others will use it—and who else runs away?"

"They may have done it to preserve what they left," suggested Halden.

"Perhaps," said Taphetta, but it was obvious he didn't think so.

The lack of air had one thing to recommend it—they needn't worry about their pests escaping. The disadvantage was that they had to wear spacesuits. They landed on top of a great building that was intact after thousands of years and still strong enough to support the added weight. And then—

Then there was nothing.



Buildings, an enormous number and variety of them, huge, not one of them less than five stories high, all with ramps instead of stairs. This was to be expected, considering the great size of the people who had lived there, and it followed the familiar pattern.

But there was nothing in those buildings! On this airless world, there was no decay, no rust or corrosion—and *nothing to decay or corrode*. No pictures, tools, nothing that resembled sculpture, and while there were places where machines had stood, none were there now. Here and there in inaccessible locations were featureless blobs of metal. The implication was clear: Where they hadn't been able to remove a machine, they had melted in down on the spot.

The thoroughness was bewildering. It wasn't done by some enemy; he would have stood off and razed the cities. But there was no rubble and the buildings were empty. The inhabitants themselves had removed all that was worth taking along.

A whole people had packed and moved away, leaving behind only massive, echoing structures.

There was plenty to learn, but nothing to learn it from. Buildings can indicate only so much and then there must be something else—at least some of the complex artifacts of a civilization—and there was none. Outside the cities, on the plains, there were the remains of plants and animals that indicated by their condition that airlessness had come suddenly. Sam Halden, the biologist, had examined them, but he discovered no clues. The unknown ancestor was still a mystery.

And the others—Emmer, the archeologist, and Meredith, the linguist—had nothing to work on, though they searched. It was Kelburn who found the first hint. Having no specific task, now that the planet was located, he wandered around in a scout ship. On the other side of the planet, he signaled that there was a machine and that it was intact!

The crew was hurriedly recalled, the equipment brought back into the ship, and they took off for the plain where Kelburn waited.

And there was the machine, immense, like everything on the planet. It stood alone, tapering toward the sky. At the base was a door, which, when open, was big enough to permit a spaceship to enter easily—only it was closed.

Kelburn stood beside the towering entrance, a tiny figure in a spacesuit. He gazed up at it as the three came near. "All we have to do is open it," he said.

"How?" asked Meredith. She seemed to have forgotten that she disliked him. He had made a chance discovery because he had nothing to do while the others were busy, but she regarded it as further proof of his superiority.

It was hard to watch the happiness that her face directed toward Kelburn. Halden turned away.

"Just press the button," he said.

Emmer noticed his expression. "It's such a big button," he objected. "It's going to be hard to know when we find it."

"There's an inscription of some sort," said Kelburn loftily. "This thing was left for a purpose. Somewhere there must be operating instructions."

"From here, it looks like a complex wave-form," a voice crinkled in their radio—Taphetta from the spaceship. "All we have to do is to find the right base in the electromagnetic spectrum and duplicate it on a beam broadcast and the door should open. You're too close to see it as clearly as I can."

Perhaps they were too close to the big ancestor, decided Halden moodily as they went back. It had overshadowed much of their thinking, and who really knew what the ancestor was like and what had motivated him?

But the Ribboneer was right about the signal, though it took several days to locate it. And then the huge door swung open and air whistled out.

Inside was another disappointment, a bare hall with a ramp leading upward, closed off at the ceiling. They could have forced through, but they had no desire to risk using a torch to penetrate the barrier—in view of the number of precautions they'd already encountered, it was logical to assume that there were more waiting for them.

It was Emmer who found the solution. "In appearance, it resembles a spaceship. Let's assume it is, minus engines. It was never intended to fly. Listen.

"There's no air, so you can't hear," said Emmer impatiently. "But you could if there were air. Put your hands against the wall."

A distinct vibration ran through the whole structure. It hadn't been there before the door opened. Some mechanism had been triggered. The rumbling went on, came to a stop, and began again. Was it some kind of communication?

Hastily rigged machines were hauled inside the chamber to generate an air supply so that sounds would be produced for the recorders. Translating equipment was set up and focused and, after some experimentation with signals, the door was slowly closed. No one remained inside; there was no guarantee that it would be as easy to get out as it had been to get in.

They waited a day and a half while the sounds were being recorded. The delay seemed endless. The happiest of the crew was Kelburn. Biologically the highest human on the expedition, he was stimulated. He wandered aimlessly and smiled affably, patting Meredith, when he came to her, in the friendliest fashion. Startled, she smiled back and looked around wanly. Halden was behind her.

If I had not been there, thought Halden—and thereafter made it a point to be there.

Meredith was excited, but not precisely happy. The work was out of her hands until the translating equipment was retrieved. As the second highest biological type, she, too, was affected, until she pointedly went to her room and locked it from the inside.

Halden kept himself awake with anti-fatigue pills, in part because Meredith could change her mind about Kelburn, and because of that locked door.

Emmer tried to be phlegmatic and seemed to succeed. Taphetta alone was unconcerned; to him, it was an interesting and perhaps profitable discovery, but important only because of that. He would not be changed at all by whatever he learned.

Hours crawled by and at last the door opened; the air came rushing out again. The translating equipment was brought back to the ship and Meredith was left alone with it.

It was half a day before she admitted the others to the laboratory.

"The machine is still working," she said. "There seems to have been some attempt to make the message hard to decode. But the methods they used were exactly the clues that the machine needed to decipher it. My function

as a linguist was to help out with the interpretation of key words and phrases. I haven't got even a little part of the message. You'll know what it is as soon as I do. After the first part, the translator didn't seem to have much trouble."

They sat down facing it—Taphetta, Kelburn, Meredith, Halden and Emmer. Meredith was midway between Kelburn and himself. Was there any significance in that, wondered Halden, or was he reading more in her behavior than was actually there?

"The translation is complete," announced the machine.

"Go ahead," Meredith ordered.

"The words will be speeded up to human tempo," said the translator. "Insofar as possible, speech mannerisms of the original will be imitated. Please remember that it is only an imitation, however."

The translator coughed, stuttered and began. "We have purposely made access to our records difficult. If you can translate this message, you'll find, at the end, instructions for reaching the rest of our culture relics. As an advanced race, you're welcome to them. We've provided a surprise for anyone else.

"For ourselves, there's nothing left but an orderly retreat to a place where we can expect to live in peace. That means leaving this Galaxy, but because of our life span, we're capable of it and we won't be followed."

Taphetta crinkled his ribbons in amusement. Kelburn frowned at the interruption, but no one else paid any attention.

The translator went on. "Our metabolic rate is the lowest of any creature we know of. We live several thousand revolutions of any recorded planet and our rate of increase is extremely low; under the most favorable circumstances, we can do no more than double our numbers in two hundred generations."

"This doesn't sound as if they were masters of biological science," rustled Taphetta.

Halden stirred uneasily. It wasn't turning out at all the way he had expected.

"At the time we left," the message continued, "we found no other intelligent race, though there were some capable of further evolution. Perhaps our scout ships long ago met your ancestors on some remote planet. We were never very numerous, and because we move and multiply so slowly, we are in danger of being swept out of existence in the foreseeable future. We prefer to leave while we can. The reason we must go developed on our own planet, deep beneath the cities, in the underworks, which we had ceased to inspect because there was no need to. This part was built to last a million generations, which is long even for us."

Emmer sat upright, annoyed at himself. "Of course! There are always sewers and I didn't think of looking there!"

"In the last several generations, we sent out four expeditions, leisurely trips because we then thought we had time to explore thoroughly. With this planet as base of operations, the successive expeditions fanned out in four directions, to cover the most representative territory."

Kelburn stiffened, mingled pride and chagrin on his face. His math had been correct, as far as he had figured it. But had there been any reason to assume that they would confine their exploration to one direction? No, they would want to cover the whole Milky Way.

Taphetta paled. Four times as many humans to contend with! He hadn't met the other three-fourths yet—and, for him, it wasn't at all a pleasant thought.

"After long preparation, we sent several ships to settle one of the nearer planets that we'd selected on the first expedition. To our dismay, we found that the plague was there—though it hadn't been on our first visit!"

Halden frowned. They were proving themselves less and less expert biologists. And this plague—there had to be a reason to leave, and sickness was as good as any—but unless he was mistaken, plague wasn't used in the strict semantic sense. It might be the fault of the translation.

"The colonists refused to settle; they came back at once and reported. We sent out our fastest ships, heavily armed. We didn't have the time to retrace our path completely, for we'd stopped at innumerable places. What we did was to check a few planets, the outward and return parts of all four voyages.

In every place, the plague was there, too, and we knew that we were responsible.

"We did what we could. Exhausting our nuclear armament, we obliterated the nearest planets on each of the four spans of our journeys."

"I *wondered* why the route came to an end," crinkled Taphetta, but there was no comment, no answer.

"We reconstructed what had happened. For a long time, the plague had lived in our sewers, subsisting on wastes. At night, because they are tiny and move exceedingly fast, they were able to make their way into our ships and were aboard on every journey. We knew they were there, but because they were so small, it was difficult to dislodge them from their nesting places. And so we tolerated their existence."

"They weren't so smart," said Taphetta. "We figured out that angle long ago. True, our ship is an exception, but we haven't landed anywhere, and won't until we deverminize it."

"We didn't guess that next to the hull in outer space and consequently exposed to hard radiation," the message went on, "those tiny creatures would mutate dangerously and escape to populate the planets we landed on. They had always been loathsome little beasts that walked instead of rolling or creeping, but now they became even more vicious, spawning explosively and fighting with the same incessant violence. They had always harbored diseases which spread to us, but now they've become hot-houses for still smaller parasites that also are able to infect us. Finally, we are now allergic to them, and when they are within miles of us, it is agony to roll or creep."

Taphetta looked around. "Who would have thought it? You were completely mistaken as to your origin." Kelburn was staring vacantly ahead, but didn't see a thing. Meredith was leaning against Halden; her eyes were closed. "The woman has finally chosen, now that she knows she was once vermin," clicked the Ribboneer. "But there are tears in her eyes."

"The intelligence of the beast has advanced slightly, though there isn't much difference between the highest and the lowest—and we've checked both

ends of all four journeys. But before, it was relatively calm and orderly. Now it is malignantly insane."

Taphetta rattled his ribbons. "Turn it off. You don't have to listen to this. We all are of some origin or other and it wasn't necessarily pretty. This being was a slug of some kind—and are you now what it describes? Perhaps mentally a little, out of pride, but the pride was false."

"We can't demolish all the planets we unthinkingly let it loose on; there are too many and it lives too fast. The stars drift and we would lose some, and before we could eliminate the last one, it would develop space travel—it has little intelligence, but it could get that far—and it would escape ahead of us. We know an impossible task when we see it. And so we're leaving, first making sure that this animal will never make use of the products of our civilization. It may reach this planet, but it will not be able to untangle our code—it's too stupid. You who will have to face it, please forgive us. It's the only thing that we're ashamed of."

"Don't listen," said the Ribboneer and, bending his broad, thin body, he sprang to the translator, shook it and banged with his ribbons until the machine was silent. "You don't have to tell anyone," crackled Taphetta. "Don't worry about me—I won't repeat it." He looked around at the faces. "But I can see that you will report to everyone exactly what you found. That pride you've developed—you'll need it."

Taphetta sat on top of the machine, looking like nothing so much as a huge fancy bow on a gift-wrapped package.

They noted the resemblance vaguely. But each of them knew that, as a member of the most numerous race in the Milky Way, no longer feared for their mysterious qualities—despised, instead—wherever they went, there would never be any gifts for them—for any man.

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